Complex Analysis

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Introduction

1.0.1 Why Complex Numbers

Chapter 2

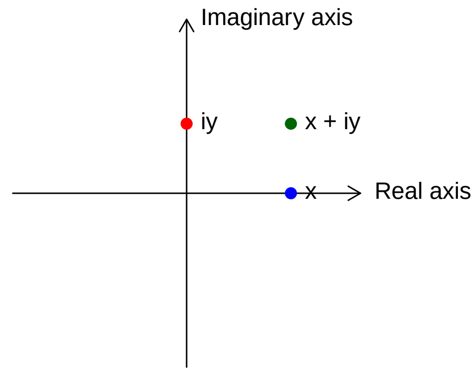
Complex Numbers

A complex number is defined as $z = x + iy$, where $x, y \in \mathbb{R}$. Here, x represents the **real part** ($\operatorname{Re} z$) and y represents the **imaginary part** ($\operatorname{Im} z$).

The set of all such numbers, $\mathbb{C}$, constitutes the complex plane. Since $\mathbb{R} \subset \mathbb{C}$, real numbers are identified as complex numbers where $\operatorname{Im} z = 0$.

There exists a bijective correspondence between $\mathbb{C}$ and the Euclidean plane $\mathbb{R}^2$, expressed as:

$$z = x + iy \longleftrightarrow (x, y)$$



In this geometric representation:

- The **real axis** corresponds to the x-axis.
- The **imaginary axis** ($i\mathbb{R}$), containing purely imaginary numbers of the form iy, corresponds to the y-axis.

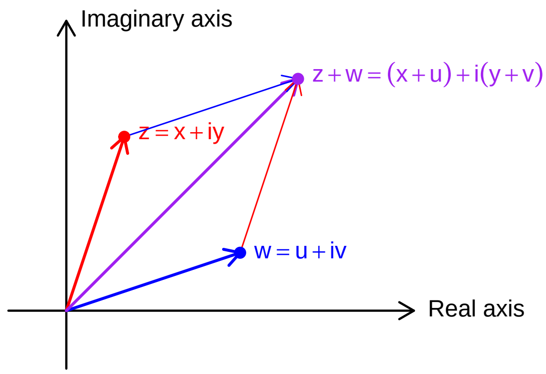
2.0.1 Basic Definitions and Algebra

- A complex number has the form $z = a + ib$ where a, b are real numbers
- a is the **real part**: $\mathrm{Re}(z) = a$
- b is the **imaginary part**: $\mathrm{Im}(z) = b$
- A number is **purely imaginary** if $a = 0$
- A number is **real** if $b = 0$
- Zero is the only number that is both real and purely imaginary

2.0.2 Arithmetic Operations

2.0.2.1 Addition

$z + w = (x + iy) + (u + iv) = (x + u) + i(y + v)$



2.0.2.2 Subtraction

$z - w = (x + iy) - (u + iv) = (x - u) + i(y - v)$

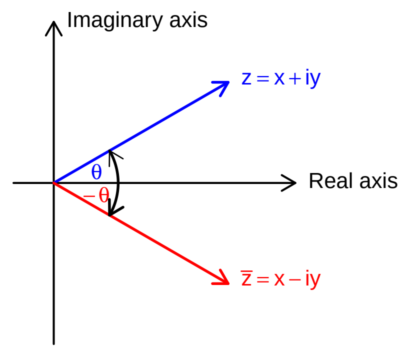
2.0.2.3 Multiplication

$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$

2.0.2.4 Division

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{ac + bd}{c^2 + d^2} + i\frac{bc - ad}{c^2 + d^2}$$

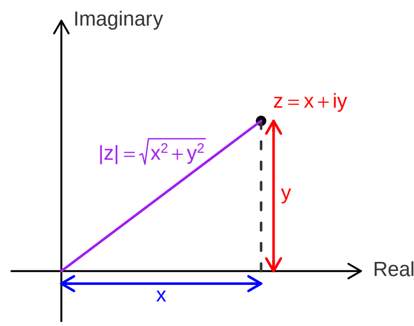
2.0.3 Complex Conjugate



- The conjugate of $z = a+ib$ is $\bar{z} = a-ib$
- Properties: For all $z, w \in \mathbb{C}$:
  - $\overline{z+w} = \bar{z} + \bar{w}$
  - $\overline{zw} = \bar{z}\,\bar{w}$
  - $\overline{(z/w)} = \frac{\bar{z}}{\bar{w}},\ w \neq 0$
  - $\frac{1}{z} = \frac{\bar{z}}{|z|^2},\ z \neq 0$
  - z is real $\iff z = \bar{z}$
  - z is purely imaginary $\iff z = -\bar{z}$

2.0.4 Absolute Value (Modulus)

- The absolute value or modulus of $z = a + ib$ is:



- $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$

- Properties:
  - $|z| \geq 0$ for all z, with equality if and only if $z = 0$
  - $|zw|^2 = (zw)(\overline{zw}) = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$

  - $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
  - $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ for $z_2 \neq 0$
  - $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)
  - $||z_1| - |z_2|| \leq |z_1 - z_2|$

2.0.5 Important Inequalities

1. **Triangle Inequality**: $|z_1 + z_2| \leq |z_1| + |z_2|$

   - Equality holds if and only if $\frac{z_1}{z_2} > 0$ (i.e., the ratio is positive real)

2. $-|z| \leq \text{Re}(z) \leq |z|$

   - Equality on right when z is positive real

3. $-|z| \leq \text{Im}(z) \leq |z|$

4. **Cauchy's Inequality**: $|a_1b_1 + a_2b_2 + ... + a_nb_n|^2 \leq (|a_1|^2 + ... + |a_n|^2)(|b_1|^2 + ... + |b_n|^2)$

   - Equality holds if and only if the a_i are proportional to the b_i

2.0.6 Fundamental Theorem of Algebra

A complex polynomial of degree $n \geq 0$ is a function of the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \qquad z \in \mathbb{C},$$

where $a_0, a_1, \ldots, a_n$ are complex numbers and $a_n \neq 0$.

A fundamental property of the complex numbers is that every polynomial with complex coefficients can be factored completely into linear factors.

Theorem 2.1 (Fundamental Theorem of Algebra)**.** *Every complex polynomial $p(z)$ of degree $n \geq 1$ admits a factorization*

$$p(z) = c\,(z - \zeta_1)^{m_1} (z - \zeta_2)^{m_2} \cdots (z - \zeta_k)^{m_k},$$

*where the numbers $\zeta_1, \ldots, \zeta_k$ are distinct complex roots of p, each multiplicity satisfies $m_j \geq 1$, and $m_1 + m_2 + \cdots + m_k = n$. This factorization is **unique**, up to a permutation of the factors.*

Proof. Omitted. Multiple complete and rigorous proofs are available at ProofWiki. □

The uniqueness of the factorization is straightforward. The points $\zeta_1, \ldots, \zeta_k$ are uniquely determined as the zeros of $p(z)$, that is, the solutions of $p(z) = 0$. Each multiplicity m_j is the unique integer $m \geq 1$ for which $p(z)$ can be written in the form $(z - \zeta_j)^m q(z)$ with $q(\zeta_j) \neq 0$.

To prove that such a factorization exists, one uses induction on the degree n of the polynomial. The key step is to show that $p(z)$ has at least one root ζ_1. Once a root is found, the polynomial can be factored as $p(z) = (z - \zeta_1) q(z)$, where $q(z)$ has degree $n-1$. By the induction hypothesis, $q(z)$ can be factored into linear factors, which completes the factorization of $p(z)$. Thus, the Fundamental Theorem of Algebra is equivalent to the statement that every complex polynomial of degree $n \geq 1$ has at least one zero.

2.1 Geometric Representation

2.1.1 The Complex Plane

- A complex number $z = a + ib$ corresponds to the point (a, b) in the plane
- The x-axis is called the **real axis**
- The y-axis is called the **imaginary axis**
- The plane is called the **complex plane**

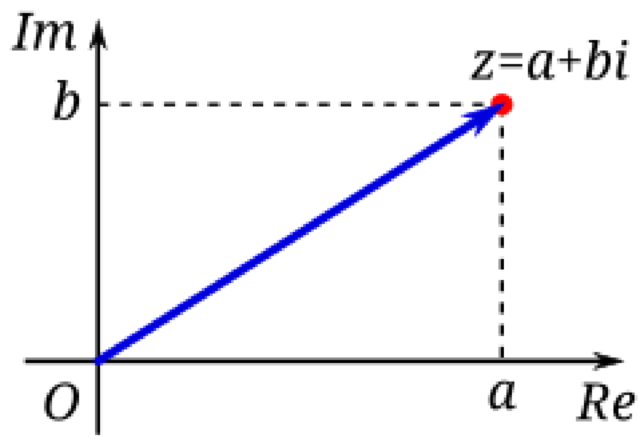


Figure 2.1: Image Source: Wikimedia

2.1.2 Vector Interpretation

- Complex numbers can be represented as vectors from the origin
- Vector addition corresponds to complex addition
- The length of the vector is $|z|$
- The distance between points z_1 and z_2 is $|z_1 - z_2|$

2.1.3 Polar Form

- A complex number can be written in **polar form**:
  - $z = r(\cos\theta + i\sin\theta)$ where $r = |z|$ and $\theta = \arg(z)$
  - r is the modulus: $r = |z| = \sqrt{a^2 + b^2}$
  - θ is the **argument** or **amplitude**: $\theta = \arg(z)$
- Properties of polar form:
  - For $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$:
  - $z_1 z_2 = r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$
  - $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
  - $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

2.1.4 De Moivre's Formula

- For $z = r(\cos\theta + i\sin\theta)$:
  - $z^n = r^n(\cos n\theta + i\sin n\theta)$
- For $r = 1$:

- $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

2.1.5 Roots of Complex Numbers

- The n-th roots of $z = r(\cos\theta + i\sin\theta)$ are:
  - $\sqrt[n]{z} = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ for $k = 0, 1, 2, \ldots, n-1$
- There are n distinct n-th roots of any non-zero complex number
- These roots have equal moduli and are equally spaced around a circle
- For $z = 1$, the roots are called the **n-th roots of unity**

2.1.6 Equations in the Complex Plane

- Circle with center a and radius r: $|z - a| = r$
  - In algebraic form: $(z - a)(\bar{z} - \bar{a}) = r^2$
- Straight line: $z = a + bt$ where t is a real parameter
  - Two lines are parallel if b' is a real multiple of b
  - Lines are orthogonal if $\frac{b'}{b}$ is purely imaginary
  - The angle between lines is $\arg \frac{b'}{b}$

2.1.7 Stereographic Projection and the Riemann Sphere

The extended complex plane is the complex plane together with the point at infinity. We denote the it by $C^* = c \cup \{\infty\}$. - The extended complex plane includes ∞ and can be represented on a unit sphere - Under stereographic projection, the point (x_1, x_2, x_3) on the sphere corresponds to: - $z = \frac{x_1 + ix_2}{1 - x_3}$

- The north pole $(0, 0, 1)$ corresponds to ∞
- Properties:
  - Circles on the sphere correspond to circles or lines in the plane
  - The distance formula: $d(z, z') = \frac{2|z-z'|}{\sqrt{(1+|z|^2)(1+|z'|^2)}}$
  - The sphere is often called the **Riemann sphere**

2.1.8 Key Formulas and Results

1. $z\bar{z} = |z|^2 = a^2 + b^2$
2. $\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{a-ib}{a^2+b^2}$
3. $z^n = r^n(\cos n\theta + i\sin n\theta)$

4. $\sqrt[n]{z} = \sqrt[n]{r}\left[\cos\left(\frac{\theta+2\pi k}{n}\right) + i\sin\left(\frac{\theta+2\pi k}{n}\right)\right]$ where $k = 0, 1, \dots, n-1$
5. Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$
6. Cauchy's inequality: $\left|\sum_{i=1}^n a_i \bar{b}_i\right|^2 \leq \left(\sum_{i=1}^n |a_i|^2\right)\left(\sum_{i=1}^n |b_i|^2\right)$
7. The equation of a circle: $|z - a| = r$
8. The roots of unity: ω^k where $\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ and $k = 0, 1, \dots, n-1$

2.2 Exercise

1. Find the values of

$$(1+2i)^3, \quad \frac{5}{-3+4i}, \quad \left(\frac{2+i}{3-2i}\right)^2, \quad (1+i)^n + (1-i)^n.$$

2. If $z = x+iy$ (where x and y are real numbers), find the real and imaginary parts of:

$$z^4, \quad \frac{1}{z}, \quad \frac{z-1}{z+1}, \quad \frac{1}{z^2}.$$

3. Show that:

$$\left(\frac{-1 \pm i\sqrt{3}}{2}\right)^3 = 1 \quad \text{and} \quad \left(\frac{\pm 1 \pm i\sqrt{3}}{2}\right)^6 = 1$$

for all combinations of signs.

Chapter 3

Green functions

Let G be a reigon and $a \in G$. Green functions G with singularity at a, the unique function $g_a : G \setminus \{0\}$

1. g_a is harmonisc on $G {a}.
2. $g_a(z) + \log(|z-a|)$ is harmonic[1] on an open neighbourhood of a.
3. $\lim_{z\to w} g_a(z) = 0 \quad \forall w \in \partial_\infty G$.

::: {.remark} $g_a(z) > 0$ for all $z \in G \setminus \{a\}$ because $g_a(z) + \log(|z-a|)$ has contious externsion to u and lim_{z\to w} g_a(z) = -\infty$ :::

We have $g_a(z)0$ on $|z-a| = r$ for r small enough (then there exist R such that its positive $\forall r < R$) Given $r < R$ such that z_0 is $|z-a| = r$. Apply max principle on $G \setminus \{z : |z-a| \leq r\}$. Since $g_a(z) \geq 0$ or $$

[1]has an exteror to u. $(a \in u)$, u is open in G.

Chapter 4

Riemann Surfaces.

Definition 4.1 (Riemann surface)**.** A **Riemann surface** is a second-countable, Hausdorff topological space X equipped with a **complex atlas** $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ satisfying the following properties:

1. Each $U_\alpha \subseteq X$ is open.
2. Each map $\varphi_\alpha : U_\alpha \to V_\alpha$ is a homeomorphism onto an open set $V_\alpha \subseteq \mathbb{C}$.
3. The collection $\{U_\alpha\}$ covers X.
4. For every pair of charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a **biholomorphism**.

Remark. **Terminology**: -"holomorphic functions" are functions which are complex differentiable on an open set in the complex plane. -A "biholomorphic map" is a holomorphic bijection whose inverse is holomorphic.

Definition 4.2. Let R_1 and R_2 be Riemann surfaces. A map $f : R_1 \to R_2$ is said to be **holomorphic** if for every pair of charts $(U, \varphi)\quad$ on $R_1, \qquad (V, \psi)\quad$ on R_2, the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic on its domain (which may be empty).

Suppose (U, φ), (U', φ') are charts on R_1, and (V, ψ), (V', ψ') are charts on R_2. On the overlap where all maps are defined, we have:

- the transition maps $\varphi' \circ \varphi^{-1}, \psi' \circ \psi^{-1}$ are **biholomorphisms** (by condition (5) in the definition of a complex atlas)

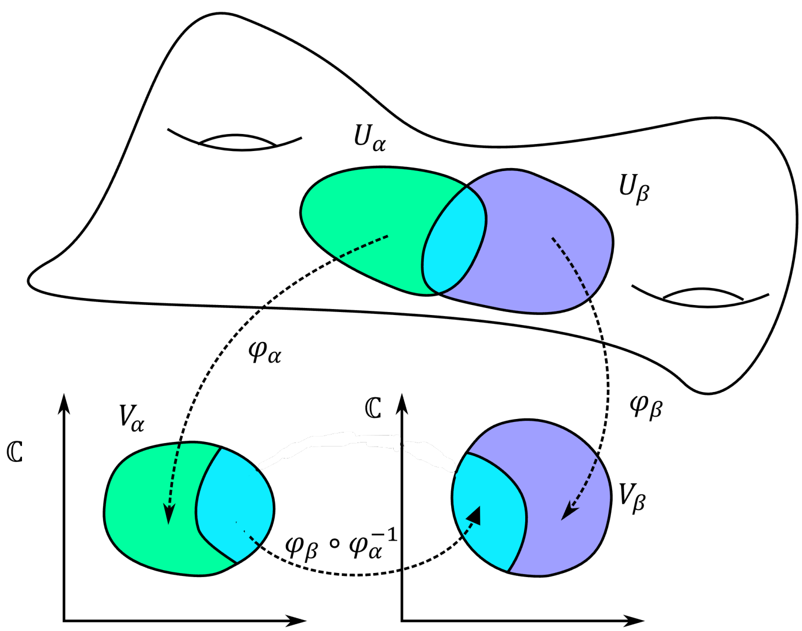


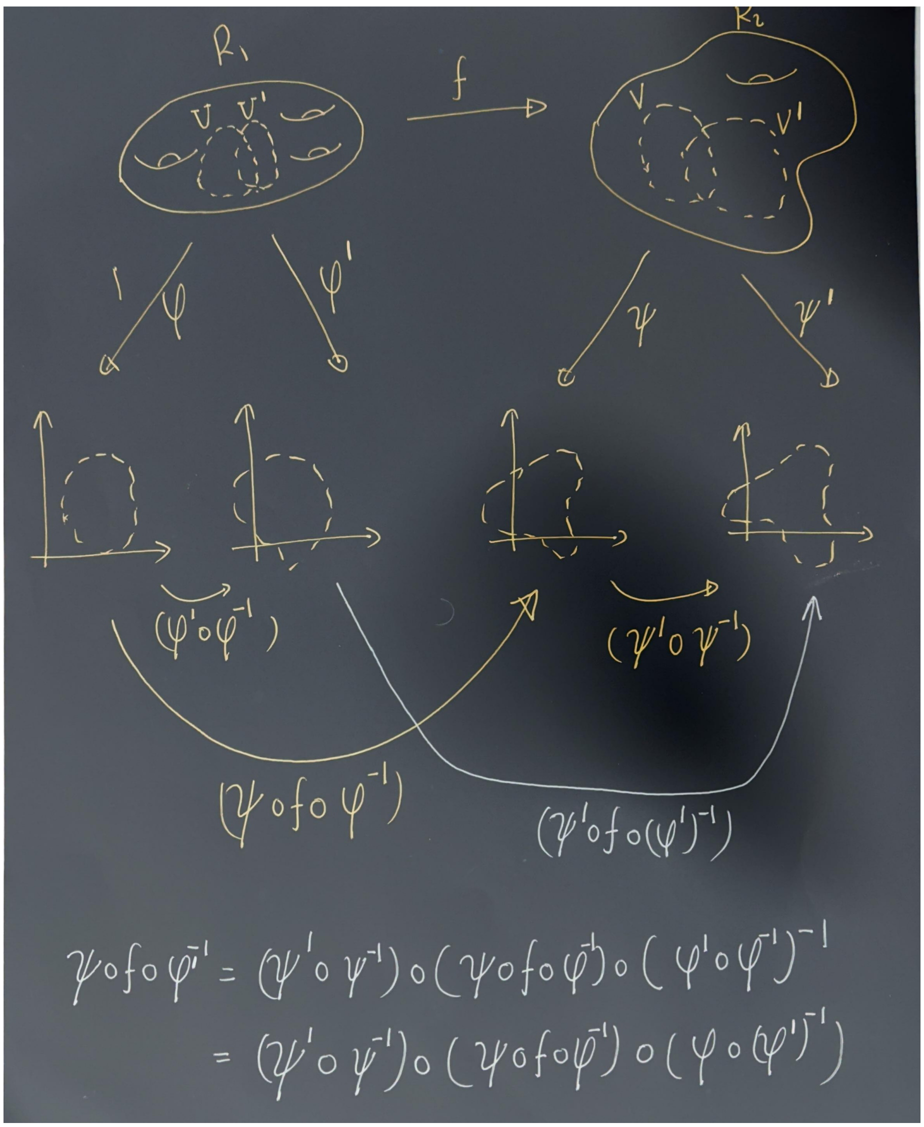
Figure 4.1: Two coordinate charts on a manifold (Image Credit: Wikipedia)

- therefore

$$\psi' \circ f \circ (\varphi')^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ (\varphi')^{-1})$$

  is holomorphic if and only if $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

Thus the notion of holomorphicity does **not** depend on the choice of charts. “‘



Exercise 4.1. Assume that for all $p \in R_1$ there is a chart (U, ϕ) such that $p \in U$ and a chart (V, ψ) such that $f(p) \in V$, so that $\psi \circ f \circ \phi^{-1}$ is holomorphic

(in the standard sense, as a complex analytic function on an open subset of the plane). Show that f is holomorphic.

Solution: We are given that for each $p \in R_1$, there exist some charts (U, ϕ) at p and (V, ψ) at $f(p)$ such that $\psi \circ f \circ \phi^{-1}$is holomorphic.

Now take any other charts (U_1, ϕ_1) at p and (V_1, ψ_1) at $f(p)$, with $f(U_1) \subset V_1$. On overlaps, the transition maps $\phi \circ \phi_1^{-1}$ and $\psi_1 \circ \psi^{-1}$ are biholomorphic (holomorphic with holomorphic inverses), because they are chart transition maps on a Riemann surface.

On the intersection where everything is defined, we can write $\psi_1 \circ f \circ \phi_1^{-1} = (\psi_1 \circ \psi^{-1}) \circ (\psi \circ f \circ \phi^{-1}) \circ (\phi \circ \phi_1^{-1})$.

- $\psi \circ f \circ \phi^{-1}$ is holomorphic by assumption.
- $\phi \circ \phi_1^{-1}$ and $\psi_1 \circ \psi^{-1}$ are holomorphic (indeed biholomorphic) transition maps.

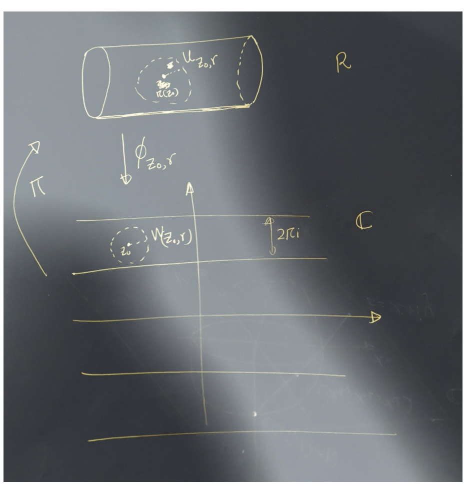
A composition of holomorphic maps is holomorphic, so $\psi_1 \circ f \circ \phi_1^{-1}$ is holomorphic.

Since this holds for any charts (U_1, ϕ_1), (V_1, ψ_1) around p and $f(p)$, it follows that f is holomorphic by the definition of a holomorphic map between Riemann surfaces.

Example 4.1. Define the equivalence relation on $\mathbb{C}$ by $z \sim w$ if and only if $z = w + 2\pi i n$ for some $n \in \mathbb{Z}$. Let $R = \mathbb{C}/\sim$ as a topological space, with projection map $\pi : \mathbb{C} \to R$.

We will define an atlas on R which will make it a Riemann surface. Define $W_{z_0,r} = \{ z \in \mathbb{C} : |z - z_0| < r \}$. Restrict to $r < \pi$ (for convenience). Let $U_{z_0,r} = \pi(W_{z_0,r})$.

Let $\varphi_{z_0,r} : U_{z_0,r} \to W_{z_0,r}$ be the unique right inverse of π on $U_{z_0,r}$ with image $W_{z_0,r}$.



Exercise 4.2. Show that for any z_1, z_2 and r_1, r_2 the map $\phi_{z_1,r_1} \circ \phi^{-1}_{z_2,r_2}$ is a translation, and in particular holomorphic. This shows that $\left\{(U_{z_0,r}, \phi_{z_0,r})\right\}_{\{z_0\in\mathbb{C},\, r>0\}}$ is an atlas making R a Riemann surface.

4.1 Solution:

4.1.1 Functions defnied on the Riemann Surface.

4.1.1.1 Exp function

We know that $\exp z$ is a holomorphic function on $\mathbb{C}$ which is periodic in the sense that $\exp(z + 2\pi in) = \exp(z)$ for all $n \in \mathbb{Z}$. Thus this passes to a well-defined function on R. Namely,

$$\mathrm{Exp} : R \longrightarrow \mathbb{C} \tag{4.1}$$
$$[z] \longmapsto \exp(z) \tag{4.2}$$

where $[z]$ is the equivalence class of z.

Exercise 4.3. Verify that this is well-defined.

Solution: To see that Exp is well-defined, assume $z \sim w$. Then there is an integer $n \in \mathbb{Z}$ such that $z = w + 2\pi i\, n$. Hence,

$$\mathrm{Exp}([z]) = e^z = e^{w+2\pi in} = e^w\, e^{2\pi in} = e^w = \mathrm{Exp}([w]).$$

Thus the value of $\mathrm{Exp}([z])$ depends only on the equivalence class $[z]$, so the function is well-defined.

Holomorphicity of Exp**.** To see that Exp is holomorphic, observe that for any chart $(U_{z_0,r}, \phi_{z_0,r})$ we have

$$\mathrm{Id} \circ \mathrm{Exp} \circ \phi^{-1}_{z_0,r}(z) = e^z,$$

which is holomorphic on $\mathbb{C}$.
Thus Exp is holomorphic by definition.

4.1.1.2 Sin function

The sine function. We may similarly define the function $\mathrm{Sin}([z]) = \sin z$, which is again well-defined and holomorphic.

Example 4.2. The complex plane $\mathbb{C}$ is a Riemann surface if we choose the atlas $\mathcal{A} = \{(U_{z_0,r}, \phi_{z_0,r})\}$. This was implicit in the previous example, where we composed by the identity map on the left.

Idea 1: Periodic functions are holomorphic functions on the Riemann surface obtained by taking quotients with respect to their periods.

Example 4.3. Let

$$R = \{(r, \theta) : r > 0,\ \theta \in \mathbb{R}\}.$$

This is trivially a topological space. We make it into a Riemann surface by defining charts as follows. For each $\theta_0 \in \mathbb{R}$, set

$$U_{\theta_0} = \{(r, \theta) : \theta_0 < \theta < \theta_0 + 2\pi,\ r > 0\},$$

and define

$$\phi_{\theta_0} : U_{\theta_0} \to \mathbb{C}, \qquad \phi_{\theta_0}(r, \theta) = re^{i\theta}.$$

Exercise 4.4. Show that for any pair θ_1, θ_2, the transition map

$$\phi_{\theta_1} \circ \phi_{\theta_2}^{-1}$$

is a biholomorphism.

4.1.1.3 LOG function

We then have a well-defined function

$$\mathrm{LOG} : R \to \mathbb{C}, \tag{4.3}$$
$$\mathrm{LOG}(r, \theta) = \log r + i\theta. \tag{4.4}$$

For any θ_0, the function

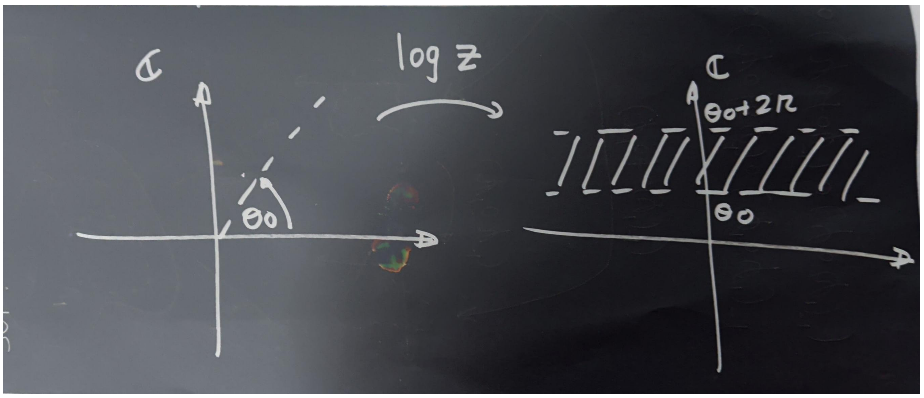
$$\mathrm{LOG} \circ \phi_{\theta_0}^{-1}$$

is a branch of the logarithm on the plane minus the slit $\arg z = \theta_0$. Explicitly,

$$\mathrm{LOG} \circ \phi_{\theta_0}^{-1}(z) = \log z,$$

where the right-hand side denotes the branch of the logarithm whose domain is the plane slit along the ray of argument θ_0, and whose values lie in the strip

$$\{w \in \mathbb{C} : \theta_0 < \Im(w) < \theta_0 + 2\pi\}.$$



4.2 The Riemann sphere

As a set, the **Riemann sphere** is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. To make this into a Riemann surface, we must specify a **topology** and an **atlas**.

A basis for the topology consists of:

1. Sets of the form $U \subset \mathbb{C}$, where U is open in $\mathbb{C}$ and $\infty \notin U$.
2. Sets of the form $\{\infty\} \cup (\mathbb{C} \setminus K)$, where $K \subset \mathbb{C}$ is compact.

Equivalently, a set $V \subset \widehat{\mathbb{C}}$ is open iff:

- If $p \in \mathbb{C}$, then there exists an open set $U \subset \mathbb{C}$ with $p \in U \subset V$.
- If $p = \infty$, then there exists a compact $K \subset \mathbb{C}$ such that $\{\infty\} \cup (\mathbb{C} \setminus K) \subset V$.

Exercise 4.5. Show that $\widehat{\mathbb{C}}$ is compact.

We define an atlas consisting of two charts:

$$U_1 = \widehat{\mathbb{C}} \setminus \{\infty\}, \qquad \varphi_1(z) = z.$$
$$U_2 = \widehat{\mathbb{C}} \setminus \{0\}, \qquad \varphi_2(z) = \frac{1}{z}.$$

The first chart and its inverse are clearly continuous.

To see that φ_2 is continuous, note that it maps $U_1 \cap U_2 = \widehat{\mathbb{C}} \setminus \{0, \infty\}$ bijectively onto $\mathbb{C} \setminus \{0\}$.

The overlap maps are:

$$\varphi_2 \circ \varphi_1^{-1}(z) = \frac{1}{z}, \qquad \varphi_1 \circ \varphi_2^{-1}(w) = \frac{1}{w}.$$

These are biholomorphic, since the inverse of $z \mapsto 1/z$ is again $z \mapsto 1/z$.

Thus, this defines a valid atlas, and the resulting Riemann surface is the **Riemann sphere**.

4.2.1 Stereographic Projection and the Metric

We identify the Riemann sphere with the unit sphere $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1\}$. Think of $\mathbb{C}$ as the plane $(x_1, y_1, 0)$. Given $z = x + iy \in \mathbb{C}$, draw the line from the north pole $(0, 0, 1)$ to $(x, y, 0)$.
This intersects the sphere at:

$$\begin{aligned} x_1 &= \frac{2x}{|z|^2 + 1} = \frac{2x}{x^2 + y^2 + 1}, \\ x_2 &= \frac{2y}{|z|^2 + 1} = \frac{2y}{x^2 + y^2 + 1}, \\ x_3 &= \frac{|z|^2 - 1}{|z|^2 + 1} = \frac{x^2 + y^2 - 1}{x^2 + y^2 + 1}. \end{aligned}$$

Define

$$\sigma : S^2 \to \widehat{\mathbb{C}}$$

by

$$\sigma(x_1, x_2, x_3) = \begin{cases} \dfrac{x_1}{1 - x_3} + i\,\dfrac{x_2}{1 - x_3}, & (x_1, x_2, x_3) \neq (0, 0, 1), \\ \infty, & (x_1, x_2, x_3) = (0, 0, 1). \end{cases}$$

$$\sigma^{-1}(z) = \begin{cases} \left(\dfrac{2\Re z}{|z|^2 + 1}, \dfrac{2\Im z}{|z|^2 + 1}, \dfrac{|z|^2 - 1}{|z|^2 + 1} \right), & z \in \mathbb{C}, \\ (0, 0, 1), & z = \infty. \end{cases}$$

The spherical (or chordal) metric is:

$$d_{\widehat{\mathbb{C}}}(z, w)| = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}},$$

with the natural extension when one point is ∞.